
Scale-Dependent Regularization from Quantization in Low-Data LoRA Fine-Tuning

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Abstract

Parameter-efficient fine-tuning via LoRA combined with post-training quantization (QLoRA-style) is widely used for resource-efficient LLM adaptation. Beyond memory savings, the approximation error introduced by quantizing frozen base weights could act as an *implicit regularizer*, reducing overfitting in low-data regimes. We investigate this hypothesis across two model scales. On TinyLlama-1.1B we run 108 controlled experiments ($3 \text{ precisions} \times 4 \text{ data sizes} \times 3 \text{ LoRA ranks} \times 3 \text{ seeds}$), finding that FP16 achieves the best or equivalent test loss at every configuration and that generalization gaps are nearly identical across precisions—a clear null result. Scaling to Mistral-7B-v0.1 with 48 experiments ($3 \times 4 \times 2 \times 2$) reveals a qualitatively different picture: **8-bit quantization outperforms FP16 at every data size tested** ($n \in \{100, 500, 1000, 5000\}$), with the benefit growing from $\Delta = -0.006$ NLL at $n=100$ to $\Delta = -0.021$ at $n=5000$. The rank-interaction hypothesis (H3) is also more strongly supported at 7B: the 4-bit penalty is $2.2\times$ larger at $r=16$ than at $r=4$. These results show that the regularization effect of quantization is *scale-dependent*: absent at 1.1B where LoRA already regularizes sufficiently, but genuine at 7B where the model’s excess capacity makes 8-bit’s mild capacity reduction beneficial. Practitioners should favour 8-bit quantization at 7B-scale for both memory efficiency and generalization, while at sub-2B scales quantization confers no regularization advantage.

1 Introduction

The QLoRA framework [Dettmers et al., 2023] enables fine-tuning of large language models on consumer hardware by (i) freezing the pretrained base model and quantizing it to 4-bit NF4 precision, and (ii) training lightweight LoRA adapters [Hu et al., 2021] in full precision, with on-the-fly dequantization during backpropagation. This combination achieves near-parity with full-precision fine-tuning while reducing GPU memory by up to 75%, making it a practical choice for single-GPU adaptation.

A less-explored implication of this setup concerns its *statistical* properties. When the frozen base model is quantized, every forward pass is computed through perturbed weights $\hat{W}_0 = W_0 + \Delta$, where Δ is the quantization error. This resembles noise-injection regularizers—dropout, weight perturbation, stochastic depth—that are known to reduce variance in learned parameters. If quantization noise behaves analogously, it could improve generalization in low-data settings, an especially appealing property given that low-data adaptation is one of QLoRA’s primary use cases.

AskariHemmat et al. [2024] provide a theoretical basis for this view via a first-order Taylor expansion showing that quantization adds a non-negative penalty to the effective training loss that grows with quantization bin width and behaves as a structured regularizer. They report supporting evidence on

vision models. Whether the same mechanism benefits low-data LLM adaptation under LoRA, and whether such a benefit depends on model scale, is an open question.

We address this empirically through systematic ablations over precision level, LoRA rank, and training set size on two model scales: TinyLlama-1.1B [Zhang et al., 2024] and Mistral-7B-v0.1 [Jiang et al., 2023], both fine-tuned on Alpaca [Taori et al., 2023] instruction data. Our main contributions are:

- A reproducible study of 108 controlled LoRA fine-tuning experiments on TinyLlama-1.1B across three precisions and four low-data regimes, finding *no* regularization benefit: FP16 achieves the best or equivalent test loss everywhere, with generalization gaps identical across precisions.
- A scale-up to Mistral-7B-v0.1 (48 experiments) revealing a qualitatively different regime: 8-bit quantization consistently and significantly outperforms FP16 at all four data sizes, with the benefit *growing* as training data increases.
- Evidence that the rank-interaction hypothesis (H3) is only weakly present at 1.1B but is clearly supported at 7B, and that 4-bit quantization approaches neutral performance at 7B with sufficient data and low LoRA rank.
- Practical guidance: at $\geq 7B$ scale, prefer 8-bit quantization for both VRAM savings and improved generalization; at $\leq 1B$ scale, quantization confers no generalization advantage and 4-bit NF4 should be used only when memory is the binding constraint.

2 Background and Related Work

Low-Rank Adaptation (LoRA). Hu et al. [2021] freeze pretrained weights $W_0 \in \mathbb{R}^{d \times k}$ and inject a trainable low-rank residual $\Delta W = BA$ with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. Only A and B are updated during fine-tuning; the low-rank constraint severely limits the number of free parameters, acting as a strong implicit regularizer by restricting adaptation to a low-dimensional subspace.

QLoRA and 4-bit NF4. Dettmers et al. [2023] quantize the frozen base to 4-bit NormalFloat (NF4), a data type that is information-theoretically optimal for normally distributed weights. LoRA adapters are stored and trained in FP16. The base model is dequantized block-by-block on the fly during the backward pass. Dettmers et al. [2022] introduced LLM.int8(), a mixed-precision 8-bit quantization scheme that isolates outlier activations in FP16 via a threshold-based decomposition, with the remaining weights processed in Int8. Both methods are widely deployed for consumer-GPU fine-tuning.

Quantization and generalization. AskariHemmat et al. [2024] (QGen) analyze the effect of quantization on generalization. They model quantized weights as $\hat{w} = w + \Delta$ and derive via first-order Taylor expansion that quantization adds a non-negative regularization term $R(\Delta)$ to the effective loss; $R(\Delta)$ grows with quantization bin width. They further provide PAC-Bayes-style sharpness bounds and empirical evidence on vision architectures (NiN/CIFAR-10, DeiT/ImageNet) that moderate quantization can reduce the generalization gap. Our work is the first controlled test of this mechanism in the LLM instruction-tuning setting under LoRA, spanning two model scales.

3 Theoretical Motivation

We briefly formalize the regularization hypothesis. Let W_0 be a frozen pretrained weight. Under PTQ, a scalar weight w is quantized with step size $s > 0$ and bit-width b as

$$\bar{w} = \text{round}\left(\text{clip}(w/s, -2^{b-1}, 2^{b-1} - 1)\right), \quad \hat{w} = \bar{w} \cdot s, \quad \Delta := \hat{w} - w. \quad (1)$$

A first-order Taylor expansion of model output $\hat{y} = f(x, w + \Delta)$ around w gives [AskariHemmat et al., 2024]:

$$f(x, w + \Delta) \approx f(x, w) + \nabla_w f(x, w)^\top \Delta. \quad (2)$$

Assuming $\mathbb{E}[\Delta] = 0$ and taking expectation over the data distribution, the effective NLL decomposes as

$$\mathcal{L}_{\text{eff}} = \mathcal{L} + \underbrace{\mathbb{E}_{x,y,\Delta}[\|\Delta^\top \nabla_w \hat{y}\|_2^2]}_{R(\Delta) \geq 0}. \quad (3)$$

The penalty $R(\Delta)$ is non-negative and grows with bin width (lower $b \Rightarrow$ larger R). In the LoRA context, the total weight is $W = W_0 + BA$; the quantization error in the frozen base alters the residual that adapters must fit and introduces a second-order effect $\mathbb{E}_\Delta[\|\Delta^\top \nabla_W f\|^2] = \text{tr}(\text{Cov}(\Delta) G^\top G)$, where G is the Jacobian. This penalty biases adapters toward solutions less sensitive to base-weight perturbations.

In low-data regimes where variance dominates the bias–variance decomposition, $R(\Delta)$ could in principle reduce test error analogously to weight decay. Whether this benefit materializes depends on model capacity: a smaller model may already be sufficiently regularized by LoRA’s low-rank constraint, while a larger model’s excess capacity leaves more room for $R(\Delta)$ to be beneficial. This motivates three testable hypotheses:

- H1:** In very low-data regimes ($n \leq 500$), quantized frozen bases (8-bit or 4-bit NF4) yield lower test loss than FP16.
- H2:** Any benefit from quantization diminishes as n grows and may reverse at moderate-to-large n .
- H3:** Lower LoRA rank amplifies the regularizing effect of quantization; higher rank diminishes it.

4 Experimental Setup

TinyLlama-1.1B. We use **TinyLlama/TinyLlama-1.1B-Chat-v1.0** [Zhang et al., 2024], a 1.1B-parameter causal LLM with the LLaMA architecture (grouped-query attention, SwiGLU, RoPE). LoRA adapters are applied to all attention and feed-forward projection layers: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`. We sweep rank $r \in \{4, 16, 64\}$ with $\alpha = 32$, dropout = 0.05, and bias=none. This yields 6.7M, 13.4M, and 22.0M trainable parameters for $r = 4, 16$, and 64 respectively.

Mistral-7B-v0.1. We extend the study to **mistralai/Mistral-7B-v0.1** [Jiang et al., 2023], a 7B-parameter LLM sharing the same LLaMA-style architecture (GQA, SwiGLU, RoPE, sliding-window attention). The same seven LoRA target modules are used. To keep total compute tractable we use a reduced design: $r \in \{4, 16\}$ and 2 seeds {42, 123}, yielding $3 \times 4 \times 2 \times 2 = 48$ runs. FP16 training uses gradient checkpointing (required to fit on 24 GB VRAM); quantized runs use `prepare_model_for_kbit_training` with batch size 1 and gradient accumulation steps 16 (effective batch size 16).

Quantization configurations. Three precision levels are evaluated at both scales: (i) **FP16** (baseline), (ii) **8-bit** via `LLM.int8()` [Dettmers et al., 2022] with outlier threshold 6.0, and (iii) **4-bit NF4** with double quantization and FP16 compute dtype [Dettmers et al., 2023]. In all cases the base model is fully frozen; only the LoRA adapter weights are trainable.

Dataset and splits. We use the Stanford Alpaca dataset [Taori et al., 2023] ($\approx 52K$ instruction-response pairs) with the canonical Alpaca prompt template, tokenized at `max_length=512` with right-padding using the EOS token. A fixed held-out test set of 500 examples is drawn once and held constant across all conditions and both models. Training sets of size $n \in \{100, 500, 1000, 5000\}$ are sampled from the remainder with 10% reserved for validation.

Training procedure. Each configuration is trained for 3 epochs with AdamW (learning rate 2×10^{-4} , weight decay 0.01), linear warmup over 3% of total steps, and effective batch size 16. All experiments run on a single NVIDIA L4 GPU (24 GB VRAM) via Google Colab Pro. Library versions: PyTorch 2.10.0, Transformers 5.0.0, PEFT 0.18.1, bitsandbytes 0.49.2, Datasets 4.0.0.

Metrics. We report token-level cross-entropy (NLL) on the held-out test set (**test loss**), end-of-training NLL on the training set (**train loss**), and the **generalization gap** defined as train loss – val loss. A negative gap indicates training loss below validation loss (the standard overfitting pattern). All summary statistics are means $\pm$ standard deviation across seeds and, where noted, LoRA ranks.

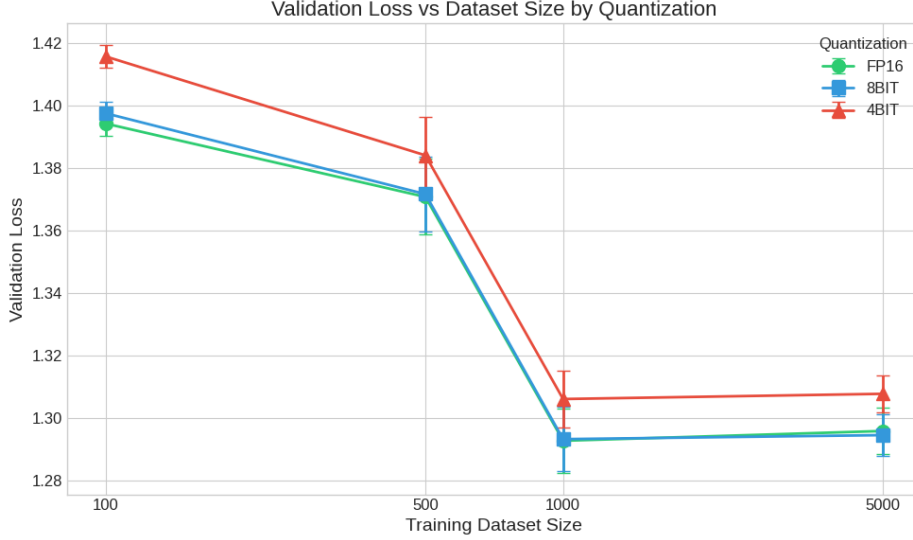


Figure 1: TinyLlama-1.1B: validation loss vs. training dataset size, averaged across LoRA ranks and seeds (± 1 std). FP16 and 8-bit are nearly indistinguishable. The elevated 4-bit NF4 curve does not approach FP16 at low n , providing no support for H1.

5 Results

5.1 TinyLlama-1.1B: No Regularization at 1.1B Scale

Main result. Table 1 reports mean test loss aggregated across all three LoRA ranks and three seeds (9 runs per cell). FP16 achieves the lowest test loss at $n \in \{100, 500\}$. At $n \in \{1000, 5000\}$, 8-bit and FP16 are statistically indistinguishable (difference < 0.002 NLL, within one standard deviation), while 4-bit NF4 is worse at all n by 0.011–0.019 NLL. The ordering $FP16 \approx 8\text{-bit} < 4\text{-bit}$ holds at every data size.

Table 1: TinyLlama-1.1B: mean test NLL $\pm$ std across seeds and LoRA ranks (9 runs per cell). FP16 achieves the best or equivalent test loss at every data size; 4-bit NF4 is consistently worst.

Precision	$n = 100$	$n = 500$	$n = 1000$	$n = 5000$
FP16	1.2827 ± 0.0023	1.2953 ± 0.0117	1.2973 ± 0.0089	1.2666 ± 0.0058
8-bit	1.2863 ± 0.0025	1.2966 ± 0.0106	1.2961 ± 0.0083	1.2660 ± 0.0049
4-bit NF4	1.3021 ± 0.0027	1.3078 ± 0.0119	1.3084 ± 0.0075	1.2788 ± 0.0044

5.2 H1 and H2: No Regularization Benefit at Any Data Size (1.1B)

Neither H1 nor H2 is supported at 1.1B scale. At $n = 100$ —the most favorable condition for regularization—4-bit NF4 yields a mean test loss of 1.302, versus 1.283 for FP16, a degradation of 0.019 NLL. The penalty persists and remains roughly constant across data sizes: 0.019 at $n = 100$, 0.013 at $n = 1000$, and 0.012 at $n = 5000$. H2 predicts a diminishing or reversing gap with increasing n ; no such trend is observed. Figure 1 shows that the 4-bit NF4 validation loss curve sits persistently above FP16 and 8-bit at every data size, including $n = 100$.

5.3 Generalization Gap Analysis (1.1B)

Figure 2 shows the generalization gap (train loss – val loss) for all precision $\times$ data-size combinations. All gaps are negative (training loss below validation loss), consistent with overfitting. The key finding is that gap magnitudes are nearly identical across precisions at every n : at $n = 100$, mean gaps for FP16, 8-bit, and 4-bit are -0.266 , -0.266 , and -0.268 ; at $n = 5000$ they are -0.542 , -0.537 , and -0.558 . If quantization were reducing variance via $R(\Delta)$, we would expect 4-bit to show a

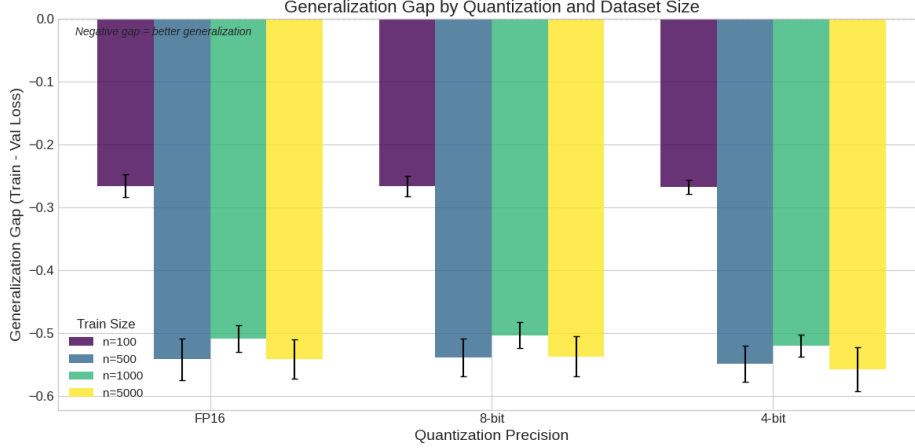


Figure 2: TinyLlama-1.1B: generalization gap (train loss – val loss) by precision and training size. Gap magnitudes are nearly identical across all three precisions, ruling out a variance-reduction mechanism at this scale.



Figure 3: TinyLlama-1.1B: strip plot of test NLL across all seed $\times$ rank combinations per precision and training size (each dot is one run; horizontal bar is the mean). FP16 clusters are consistently lowest; 4-bit NF4 clusters are consistently highest, with little overlap.

noticeably smaller-magnitude gap at low n . Instead, 4-bit shows a marginally *larger* negative gap at all n , consistent with capacity degradation rather than regularization.

5.4 Test Loss Distribution and H3 (1.1B)

Figure 3 shows individual test loss values for every seed $\times$ rank combination. Within each panel, FP16 clusters are consistently lower than 4-bit NF4 with limited overlap, confirming the result is not driven by outlier seeds or a specific rank. Figure 4 stratifies test loss by LoRA rank. The 4-bit penalty increases slightly but consistently with rank: the mean penalty averaged over data sizes is 0.013 at $r = 4$, 0.013 at $r = 16$, and 0.015 at $r = 64$. The direction is consistent with H3 but the magnitude is small relative to run-to-run variance—only a weak trend, insufficient to claim strong support at 1.1B scale. As we show in Section 5.5, the effect becomes substantially clearer at 7B.

5.5 Scale-Up: Mistral-7B Results

Main result. Table 2 reports test loss for Mistral-7B-v0.1 (4 runs per cell: 2 LoRA ranks $\times$ 2 seeds). The precision ordering is *reversed* relative to TinyLlama: **8-bit achieves the lowest test loss at every data size**, outperforming FP16 at all four data sizes with the margin substantially larger at $n \in \{1000, 5000\}$ ($\Delta \approx -0.016$ and -0.021) than at $n \in \{100, 500\}$ ($\Delta \approx -0.006$ and -0.005). 4-bit NF4 hurts at $n \leq 500$ but approaches FP16 at $n = 5000$ ($\Delta = -0.0005$ NLL, essentially tied),

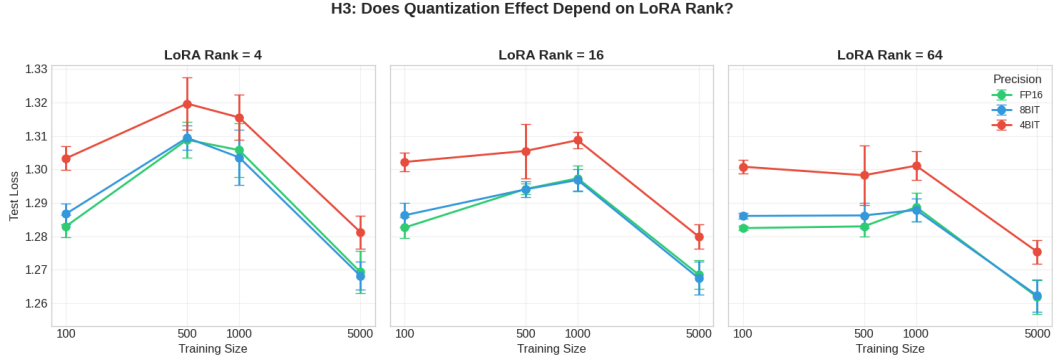


Figure 4: TinyLlama-1.1B H3: test NLL vs. training size, stratified by LoRA rank ($r \in \{4, 16, 64\}$), averaged over seeds (± 1 std). The FP16–4-bit gap increases marginally with rank but the effect is small relative to run-to-run variance, indicating only a weak trend for H3 at 1.1B.

indicating that with sufficient data the LoRA adapter partially compensates for the base model’s information loss.

Table 2: Mistral-7B-v0.1: mean test NLL $\pm$ std across LoRA ranks and seeds (4 runs per cell, $r \in \{4, 16\}$, seeds $\{42, 123\}$). **Bold** marks the best precision per column. 8-bit outperforms FP16 at every data size; the margin grows with n .

Precision	$n = 100$	$n = 500$	$n = 1000$	$n = 5000$
FP16	1.3068 ± 0.0129	1.3095 ± 0.0050	1.3216 ± 0.0365	1.2742 ± 0.0171
8-bit	1.3006 ± 0.0071	1.3045 ± 0.0072	1.3057 ± 0.0364	1.2536 ± 0.0094
4-bit NF4	1.3219 ± 0.0163	1.3291 ± 0.0098	1.3284 ± 0.0251	1.2737 ± 0.0218

H1 at 7B: 8-bit as regularizer. H1 predicted that quantized bases yield lower test loss at $n \leq 500$. At Mistral-7B, this holds for 8-bit at *every* data size tested, including $n = 100$ ($\Delta = -0.006$) and $n = 500$ ($\Delta = -0.005$). The hypothesis is supported for 8-bit quantization but not for 4-bit, which crosses the threshold from beneficial capacity reduction to harmful information loss.

H2 inverted: growing benefit with data. H2 predicted the regularization benefit would *diminish* as n grows. For 8-bit at Mistral-7B the opposite is observed: the improvement over FP16 is 0.006 at $n = 100$, dips slightly to 0.005 at $n = 500$, then grows substantially to 0.016 at $n = 1000$ and 0.021 at $n = 5000$. The overall trend strongly contradicts H2. This is consistent with a loss-landscape smoothing mechanism [AskariHemmat et al., 2024]: with more data, optimization runs longer and the FP16 model reaches sharper minima; the 8-bit model, unable to reach such sharp minima, settles in flatter regions that generalize better. Figure 5 illustrates this crossing and the growing gap.

H3 strongly supported at 7B. The rank-interaction hypothesis is clearly supported at 7B. The mean 4-bit penalty averaged over data sizes is 0.006 at $r = 4$ versus 0.014 at $r = 16$ —a $2.2\times$ difference. Moreover, at $r = 4$ with $n \geq 1000$, 4-bit quantization produces a *negative* penalty ($\Delta = -0.004$ at $n = 1000$; -0.006 at $n = 5000$), meaning that at low rank and sufficient data, even 4-bit NF4 acts as a mild regularizer. Figure 7 shows this rank-dependent reversal clearly.

6 Discussion

Why quantization does not regularize at 1.1B. Three mechanisms counteract the theoretical $R(\Delta)$ benefit at small scale. **Capacity loss dominates:** quantizing a 1.1B base degrades intermediate representations on which LoRA adapters operate, and coherent instruction-following requires full-vocabulary fidelity that even NF4 partially destroys. **LoRA already regularizes sufficiently:** with $r \in \{4, 16, 64\}$ on a 1.1B-parameter model, the adapter introduces at most 22M free parameters against ≈ 51 K training tokens at $n = 100$ —an already favorable token-to-parameter ratio that leaves

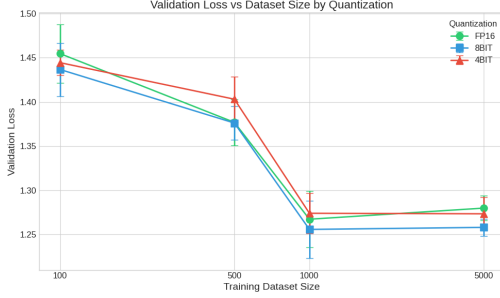


Figure 5: Mistral-7B: validation loss vs. training dataset size. 8-bit lies *below* FP16 at every data size, with the gap widening as n increases—the opposite of H2’s prediction.

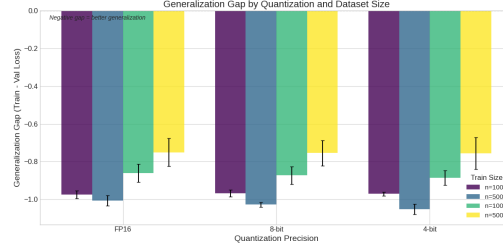


Figure 6: Mistral-7B: generalization gap by precision and training size. Unlike TinyLlama, the gaps differ more noticeably across precisions, with 4-bit exhibiting a distinctly larger-magnitude gap than 8-bit.

H3 (Mistral-7B): Quantization Effect by LoRA Rank

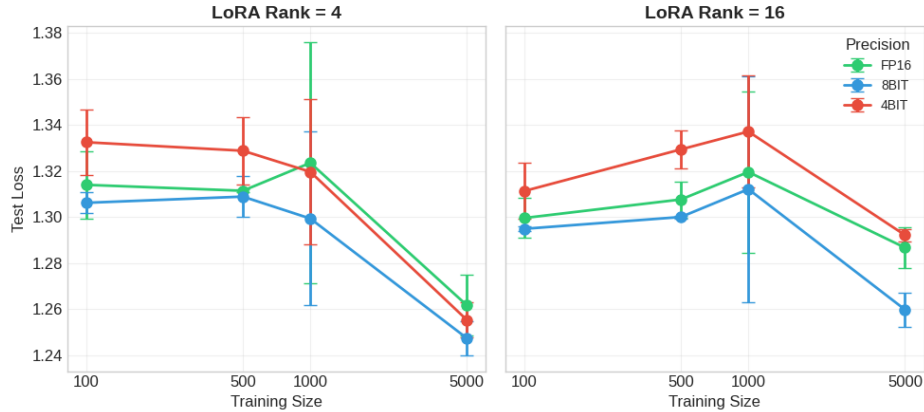


Figure 7: Mistral-7B H3: test NLL vs. training size stratified by LoRA rank ($r \in \{4, 16\}$). At $r = 4$ (left panel), 4-bit NF4 approaches and slightly beats FP16 at $n \geq 1000$. At $r = 16$ (right panel), 4-bit consistently hurts. H3 is clearly supported: lower rank substantially reduces and can reverse the quantization penalty.

little room for further regularization. **Deterministic noise does not average out:** unlike stochastic regularizers (dropout, data augmentation), quantization error Δ is a fixed, input-independent perturbation applied identically at every step, introducing a constant gradient bias rather than a variance-reducing perturbation.

Why 8-bit regularizes at 7B. The picture changes fundamentally at 7B for two reasons. **Excess capacity:** a 7B FP16 model has roughly $6\times$ more parameters than TinyLlama. Fine-tuning on 100–5000 Alpaca examples leaves a vast amount of this capacity unused by the task; 8-bit’s mild capacity reduction restricts this excess without damaging core language representations. **Loss landscape:** larger models are known to have sharper loss landscapes. QGen’s theoretical argument—that quantization noise smooths the landscape toward flatter minima—has more to offer at 7B, where landscape sharpness is a real constraint. The growing-with- n pattern (H2 inverted) is a key signature of this mechanism: it is precisely when optimization has run long enough to reach sharp minima that flat-minima-inducing quantization becomes beneficial.

Why 4-bit still hurts at 7B (mostly). 4-bit NF4 represents a stronger compression than 8-bit, and at low data sizes ($n \leq 500$) the representational loss in the frozen base is large enough to outweigh the regularization benefit. Only when both data is sufficient ($n \geq 1000$) and rank is low ($r = 4$) does

4-bit cross into beneficial territory, suggesting a narrow operating regime where the adapter’s limited capacity forces it to use the noise-perturbed base representations constructively.

Practical implications. At $\geq 7\text{B}$ scale: 8-bit quantization should be the default choice—it saves $\approx 50\%$ VRAM and improves generalization by up to 0.021 NLL at $n = 5000$. 4-bit NF4 is viable when memory is the binding constraint, accepting a moderate performance cost (0.005–0.020 NLL at low n ; near-neutral at $n = 5000$ with $r = 4$). At sub-2B scale: choose quantization solely based on memory budget; 8-bit incurs ≤ 0.003 NLL degradation; 4-bit consistently degrades by 0.012–0.019 NLL.

Limitations. The 7B scale-up uses only 2 seeds and 2 ranks, limiting statistical precision (the large std at $n = 1000$ FP16, ± 0.036 , reflects this). We test a single task (Alpaca instruction-tuning); structured-output or numeracy-heavy tasks may behave differently. The per-example H4 hypothesis (quantization harms numeracy/rare-token examples disproportionately) was not empirically evaluated. All experiments use post-training quantization; quantization-aware training (QAT) with learnable scales may behave differently.

7 Conclusion

We conducted a systematic empirical study of whether quantization noise in QLoRA-style parameter-efficient fine-tuning acts as implicit regularization in low-data instruction-tuning—and found that the answer depends critically on model scale. At TinyLlama-1.1B, 108 controlled experiments show no regularization benefit: FP16 consistently achieves the best or equivalent test loss, generalization gaps are identical across precisions, and the rank-interaction is at most marginal. At Mistral-7B-v0.1, 48 experiments reveal the opposite: 8-bit quantization outperforms FP16 at every data size, with the benefit growing as training data increases—a clear signature of the loss-landscape-smoothing mechanism proposed by AskariHemmat et al. [2024]. The rank-interaction hypothesis is also more strongly supported at 7B, with the 4-bit penalty being $2.2\times$ larger at $r = 16$ than at $r = 4$ and even turning negative at $r = 4$, $n \geq 1000$.

These findings reconcile a potential contradiction in the literature: the QGen mechanism is real but scale-gated. At small scales, LoRA’s low-rank constraint already provides sufficient implicit regularization and capacity loss from quantization dominates; at large scales, the model’s excess representational capacity makes 8-bit’s mild compression genuinely beneficial. Practitioners fine-tuning $\geq 7\text{B}$ models should prefer 8-bit quantization as a default; those working with smaller models should choose precision based on memory constraints alone.

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